As Linda and Bret took turns driving, I read and thought. (Linda and Bret were accustomed to such introversion; they had lived with me for many years.) The novel was fun, but Carl, indeed, was in trouble. He had his heroine, Eleanor Arroway, plunge into a black hole near Earth, travel through hyperspace in the manner of Figure 13.4, and emerge an hour later near the star Vega, 26 light-years away. Carl, not being a relativity expert, was unfamiliar with the message of perturbation calculations[2]: *It is impossible to travel through hyperspace from a black hole's core to another part of our Universe.* Any black hole is continually being bombarded by tiny electromagnetic vacuum fluctuations and by tiny amounts of radiation. As these fluctuations and radiation fall into the hole, they get accelerated by the hole's gravity to enormous energy, and they then rain down explosively on any "little closed universe" or "tunnel" or other vehicle by which one might try to launch the trip through hyperspace. The calculations were unequivocal; any vehicle for hyperspace travel gets destroyed by the explosive "rain" before the trip can be launched. Carl's novel had to be changed.

During the return drive from Santa Cruz, somewhere west of Fresno on Interstate 5, a glimmer of an idea came to me. Maybe Carl could replace his black hole by a *wormhole* through hyperspace.

A wormhole is a hypothetical shortcut for travel between distant points in the Universe. The wormhole has two entrances called "mouths," one (for example) near Earth, and the other (for example) in orbit around Vega, 26 light-years away. The mouths are connected to each other by a tunnel through hyperspace (the wormhole) that might be only a kilometer long. If we enter the near-Earth mouth, we find ourselves in the tunnel. By traveling just one kilometer down the tunnel we reach the other mouth and emerge near Vega, 26 light-years away as measured in the external Universe.

Figure 14.1 depicts such a wormhole in an embedding diagram. This diagram, as is usual for embedding diagrams, idealizes our Universe as having only two spatial dimensions rather than three (see Figures 3.2 and 3.3). In the diagram the space of our Universe is depicted as a two-dimensional sheet. Just as an ant crawling over a sheet of paper is oblivious to whether the paper is lying flat or is gently folded, so we in our Universe are oblivious to whether our Universe is lying flat in hyperspace or is gently